(2) $y''(t)+4y'(t)+4y(t)=f'(t)+3f(t)$，$y(0_-)=1$，$y'(0_-)=2$，$f(t)=e^{-t}\varepsilon(t)$。

解: 零输入响应:

$y_{zi}''(t)+4y_{zi}'(t)+4y_{zi}(t)=0$

$r^2+4r+4=0$

$r_1=r_2=-2$

$\therefore y_{zi}(t)=(C_1+C_2x)e^{-2x}$

$y_{zi}(0^+)=y_{zi}(0^-)=y_A(0^-)=1$

$y_{zi}'(0_+)=y_{zi}'(0_-)=y_A(0_-)=2$

$\therefore C_1=1$

$C_2-2C_1=2$

$\therefore C_1=1,\ C_2=4$

$\therefore y_{zi}(t)=(C_1+4x)e^{-2x}$

零状态响应

$y_{zs}''(t)+4y_{zs}'(t)+4y_{zs}(t)=-e^{-t}\varepsilon(t)+e^{-t}\delta(t)+3e^{-t}\varepsilon(t)$

$y_{zs}''(t)+4y_{zs}'(t)+4y_{zs}(t)=2e^{-t}\varepsilon(t)+e^{-t}\delta(t)$

求 $y_{zs}(0^+)$与$y_{zs}(0^-)$ 系数匹配

$\therefore \int_{0^-}^{0^+} y_{zs}''(t)dt+4\int_{0^-}^{0^+} y_{zs}'(t)dt+4y_{zs}(t)dt=2\int_{0^-}^{0^+} e^{-t}\varepsilon(t)dt+\int_{0^-}^{0^+} e^{-t}\delta(t)dt$

$\because y_{zs}''(t)$中有$\delta(t)$

$\therefore y_{zs}'(0^+)-y_{zs}'(0^-)=1$

$y_{zs}(0^+)-y_{zs}(0^-)=0$

$\therefore y_{zs}'(0^+)=1$

$y_{zs}(0^-)=0$

$t>0$时

$y_{zs}''(t)+4y_{zs}'(t)+4y_{zs}(t)=2e^{-t}$

先求特解为: $2e^{-t}$

$\therefore y_{zs}(t)=(C_1+C_2t)e^{-2t}+2e^{-t}$

$\therefore C_1+2=0$

$C_2-2C_2-2=1$

$\therefore C_1=-2,\ C_2=-1$

$\therefore y_{zs}(t)=(-2-t)e^{-2t}+2e^{-t}$

$\therefore \cancel{y_{zs}(t)=(-1-2t)e^{-2t}+2e^{-t}}$